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SentiFive: A Multi-Class Bengali Dataset for Sentiment Analysis IEEE

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***Abstract*—Sentiment Analysis (SA) has become an essential task in natural language processing (NLP), enabling computational understanding of opinions expressed in text. Although Bengali ranks among the most spoken languages globally, it remains significantly underrepresented in sentiment analysis research. To address the scarcity of rich and fine-grained Bengali datasets, we present SentiFive a large-scale, manually annotated dataset consisting of 31,411 Bengali YouTube comments, labeled across five sentiment categories: নিশ্চিত নেতিবাচক (Strongly Negative), কিছুটা নেতিবাচক (Slightly Negative), নিরপেক্ষ (Neutral), নিশ্চিত ইতিবাচক (Strongly Positive), and কিছুটা ইতিবাচক (Slightly Positive). Over 400 native Bengali speakers contributed to the rigorous annotation process, ensuring reliability and diversity in sentiment representation. We benchmarked the dataset across binary, three-class, and five-class classification tasks using both traditional machine learning (ML) models and deep learning (DL) architectures. Experimental results demonstrate that performance declines as class granularity increases, with LSTM and BiLSTM models consistently outperforming ML baselines. Notably, LSTM achieved 74% accuracy for binary classification, BiLSTM achieved 59% for three-class, and LSTM achieved 39% for five-class classification. This work provides a foundational resource for advancing Bengali NLP and highlights the challenges and potential in fine-grained sentiment modeling for low-resource languages. We established performance baselines and intend to make the dataset "SentiFive" and the related code-base publicly available for future research.This work provides a foundational resource for advancing Bengali NLP [1].**

***Index Terms*—SentiFive, 5-Class Sentiment, Baseline Evaluation, LSTM, BiLSTM, Bangla Natural Language Processing (BNLP), Sentiment Analysis (SA)**

I. Introduction

Sentiment analysis (SA), also known as opinion mining, is a key area of Natural Language Processing (NLP) that focuses on identifying opinions, emotions, and attitudes expressed in text [1]. SA can be performed at multiple levels: document-level, sentence-level, and aspect-level. While document-level analysis provides an overall sentiment of a text, sentence-level analysis examines individual sentences, and aspect-level analysis identifies sentiments related to specific features.

Bangla sentiment analysis is an active research area in Natural Language Processing (NLP) since Bangla is the seventh most spoken language worldwide, with about 273 million speakers [2]. However, SA in Bangla remains underexplored, especially for fine-grained, multi-class classification. Existing datasets vary widely in size and number of classes. Many are small or limited to binary and three-class settings, while others are large but highly skewed toward specific sentiments. These limitations restrict the development of robust models for nuanced sentiment understanding. A detailed analysis of existing datasets is provided in Section II.

To overcome these issues, this study introduces SentiFive, a large-scale Bangla sentiment dataset containing 31,411 YouTube comments annotated into five sentiment classes: Strongly Negative, Slightly Negative, Neutral, Slightly Positive, and Strongly Positive. The dataset offers a more balanced and fine-grained structure, supporting effective multi-class sentiment modeling.

The contributions of this study are as follows:

- Creation of SentiFive, a large, manually annotated Bangla sentiment dataset with five classes, addressing limitations of existing multi-class datasets.
- Established baseline performance for binary, three-class, and five-class sentiment classification tasks using a variety of ML and DL models.
- Highlighting the performance decline as the number of classes increase with DL models outperforming the traditional ML approaches.

II. Literature Review

Numerous researchers have explored Bengali sentiment analysis from different angles. The authors of [3] examined 5,500 book reviews using four machine learning algorithms and reported that the random forest model delivered the strongest performance for sentiment polarity classification. Similarly, [1] introduced BANGLABOOK, a large-scale dataset of 158,065 Bangla book reviews labeled as positive, negative, or neutral. Their work showed that pre-trained models consistently outperform approaches based on manually engineered features, emphasizing the ongoing need for more extensive training resources in this field.

Islam et al. [4] introduced SentNoB, a dataset of informal Bangla text built from public social-media comments

[1]Code: https://github.com/ARASHFAQUE/SentiFive-Code Dataset: https://www.kaggle.com/datasets/amermahbub01/sentifive/data

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across 13 domains. The data is labeled for sentiment polarity (positive, negative, neutral) and reflects a wide range of dialectal variation and grammatical irregularities. Funded by Bangladesh govt [5] introduced Sentigold, which improves upon SentNoB by offering a larger dataset (70,000 vs. 15,000 samples) and more nuanced sentiment classifications with five distinct classes across 30 domains. Unlike SentNoB, which has labeling errors and a low IAA score of 0.53. Additionally, SentiGOLD covers more diverse domains, enhancing its generalizability for real-world sentiment analysis.

Emotion classification, a branch of sentiment analysis, aims to identify discrete emotional states rather than general sentiment polarity. In Bangla NLP, several multi-class emotion datasets have been developed to advance affective text understanding. For example, the Unified Bangla Multi-Class Emotion Corpus (UBMEC) combines BNEmo (6,327 Facebook comments), BEmoC (7,000 texts from social media, blogs, novels, and newspapers), and manually annotated reviews from YouTube and BBC Bangla, totaling 13,072 reviews across six emotion classes [6]. Another effort [7] combined BanglaEmotion (6,314 Facebook comments) and BEmoC (7,000 texts) to produce 12,628 labeled instances across six emotion categories using LR, MNB, and MLP models with TF-IDF and CountVectorizer features. Additionally, [8] constructed a Bangla emotion corpus by machine-translating 7,214 English sentences from a Kaggle dataset using Google Translate. Unprocessed or inaccurately translated samples were retranslated iteratively. The finalized corpus was annotated with six distinct emotion categories.

While these works significantly advance sentiment and emotion analysis in Bangla, several limitations remain in existing datasets. Many multi-class datasets are relatively small and exhibit imbalance in certain classes. For example, Ahmed et al. [9] and Rayhan et al. [8] cover six sentiment classes, but minority classes such as Surprise are severely underrepresented. Even moderately larger datasets like Saurav et al. [6]remain limited in size for robust multi-class modeling. On the other hand, some large datasets, such as Kabir et al. [1] and Rashid et al. [10], are highly skewed, favoring specific sentiments, which can bias model learning. Binary and three-class datasets, such as those of Hassan et al. [11] and Junaid et al. [11] are balanced but too small to support complex multi-class classification. While SentiGOLD represents a major advancement in developing a standardized Bangla sentiment resource, it is primarily composed of data from 30 curated domains, collected and annotated under expert supervision following strict linguistic guidelines. In contrast, our **SentiFive** dataset is constructed from user-generated social media content which tend to be informal, noisy, and rich in colloquial expressions. Unlike the controlled and domain-specific nature of SentiGOLD, SentiFive captures the spontaneity and diversity of real-world Bangla sentiment as expressed in online discourse. Furthermore, while SentiGOLD relies on a limited number of expert annotators, SentiFive employs a large-scale community annotation process involving over 400 contributors, ensuring broader linguistic and perceptual coverage. These distinctions in data source, annotation methodology, and linguistic diversity collectively make SentiFive a distinct resource for modeling sentiment in naturally occurring Bangla text. By combining sufficient data size, balanced class distribution, and naturally occurring social media language, SentiFive supports more nuanced sentiment analysis and mitigates issues of small, skewed, or underrepresented classes observed in prior work.

Name	**Size**	**Classes**	**Class Distribution**	**Comment**
Ahmed et al. [7]	12,628	6	Sad 2668 Happy 3432 disgust 2059 Surprise 1341 fear 1346 angry 2468	Medium-sized, class imbalance due to smaller surprise class.
Rayhan et al. [8]	7,214	6	Happy 1362 Sad 1352 Fear 1395 Angry 1373 Love 1199 Surprise 533	Medium-sized, class imbalance due to smaller surprise class.
Sourav et al. [6]	13,072	6	Joy 3290 Sadness 2622 Anger 2422 Disgust 2049 Fear 1348 Surprise 1341	Six-class, multi-domain Bangla dataset, moderately balanced.
Islam et al. [12]	44,491	3	Positive 14424 Negative 12371 Neutral 17696	Large dataset, relatively balanced classes.
Hassan et al. [11]	1,141	2	Positive 570 Negative 571	Small and balanced dataset.
Kabir et al. [1]	158,065	3	Positive 141587 Negative 6946 Neutral 6804	Large but highly imbalanced.
Junaid et al. [13]	1,040	2	Positive 520 Negative 520	Small and perfectly balanced.
Alvi et al. [14]	7,000	2	Positive 3500 Negative 3500	Moderate-sized, balanced.
Rashid et al. [10]	78,130	2	Positive 67268 Negative 10862	Large, skewed to positive.
SentiFive (Proposed)	31,411	5	Strongly Negative 6780 Slightly Negative 6006 Neutral 6453 Slightly Positive 5635 Strongly Positive 6537	Five-class, multi-domain Bangla dataset, balanced.

TABLE I: Existing Sentiment Analysis Datasets for Bangla

III. Dataset Preparation

We chose YouTube as our data source to filter out relevant user-generated comments. We developed a Python script to collect comments from YouTube videos. To ensure privacy and maintain the anonymity of the commenters, our script extracted only the comments, excluding any user information such as usernames, profile details, or other identifying data.

The comments were carefully selected from a wide variety of well-known YouTube channels, ensuring a wide diversity of subjects and opinions. Extraction was then followed by verifying with human judgments to filter out spam content, keeping the quality intact by ensuring the integrity of the dataset.

A. Data Annotation

More than 400 students were involved in the annotation process, with all of them being students of the university from the first year to the final year. Each sentence was labeled at least 5 times by different students to ensure correct labeling. We categorized sentiments into five distinct groups: নিশ্চিত নেতিবাচক (Strongly Negative), কিছুটা নেতিবাচক (Slightly Negative), নিরপেক্ষ (Neutral), নিশ্চিত ইতিবাচক (Strongly Positive), and কিছুটা ইতিবাচক (Slightly Positive). After the first round of annotation, the most frequent sentiment was selected as the final sentiment for each sentence. When all five annotators assigned different labels, or when a tie occurred among the most frequent labels, the instance was re-examined by the authors, who collaboratively discussed and determined the final sentiment label by consensus to ensure consistency and annotation reliability. Table II shows the data annotation process for some data.

Text Data	আবার কতগুলো গাধা এখানে এসে কমেন্ট করেছে। (Some donkeys have come here and commented again.)	বস এগিয়ে যান আমরা সাথে আছি (Boss, move forward; we are with you.)	ভাই রাজস্থানের খাবারের রিভিউ দিয়েন (Brother, give a review of Rajasthan's food.)
1st annotator	নিরপেক্ষ (Neutral)	নিশ্চিত ইতিবাচক (Strongly Positive)	নিরপেক্ষ (Neutral)
2nd annotator	নিশ্চিত নেতিবাচক (Strongly Negative)	নিশ্চিত ইতিবাচক (Strongly Positive)	নিশ্চিত ইতিবাচক (Strongly Positive)
3rd annotator	নিশ্চিত নেতিবাচক (Strongly Negative)	নিশ্চিত ইতিবাচক (Strongly Positive)	নিশ্চিত ইতিবাচক (Strongly Positive)
4th annotator	নিশ্চিত নেতিবাচক (Strongly Negative)	নিশ্চিত ইতিবাচক (Strongly Positive)	নিরপেক্ষ (Neutral)
5th annotator	নিশ্চিত নেতিবাচক (Strongly Negative)	নিশ্চিত ইতিবাচক (Strongly Positive)	নিরপেক্ষ (Neutral)
Final Label	নিশ্চিত নেতিবাচক (Strongly Negative)	নিশ্চিত ইতিবাচক (Strongly Positive)	নিরপেক্ষ (Neutral)

TABLE II: Data Annotation Process

B. Data Processing

For data processing, we cleaned the dataset by removing non-Bengali words, punctuation, emojis, hashtags, and other irrelevant elements.

1) Punctuation, Non-Bangla Words, and Emoji Removal: When people express their opinions through comments, they sometimes use punctuation or emojis to embody their actual emotions; for example, "বাঙালি চিরন্তন ❤" here, the commenter is trying to say "Bengali is eternal" to express their admiration and added a heart to emphasize their feelings. Removing punctuation (e.g., "?", "!", "-", ",", and so on), non-Bangla words and emojis from our dataset was quite an important step. Table III shows the dataset cleaning procedure. Table III shows the dataset cleaning procedure.

Before Cleaning	**After Cleaning**
ক্যামেরা অতিরিক্ত সাদা হয়ে যাচ্ছিলো, প্লেসমেন্ট ও ঠিক নাই (The camera was becoming excessively white, and the placement is also not correct.)	ক্যামেরা অতিরিক্ত সাদা হয়ে যাচ্ছিলো প্লেসমেন্ট ও ঠিক নাই (The camera was becoming excessively white, and the placement is also not correct.)
মায়েরা সব জায়গায় একিরকম হয় ❤❤❤ (Mothers are the same everywhere ❤❤❤)	মায়েরা সব জায়গায় একিরকম হয় (Mothers are the same everywhere)

TABLE III: Dataset Cleaning Process

2) Tokenization: Tokenization involves dividing each sentence into individual words or tokens, which allows for a more thorough analysis of the text. For example, the comment "ছাত্র রাজনীতি বন্ধ করা হোক" [Let student politics be banned] after tokenizing becomes: "ছাত্র" (Student), "রাজনীতি" (politics), "বন্ধ" (banned), "করা" (to be), "হোক" (let).

C. Dataset Observation

1) Balanced Dataset: When a class has a significantly higher number of samples than other classes, it is called the "class imbalance." Because of the biases towards the majority class, this imbalance can degrade the performance of the system and thus struggle to identify the minority classes. Our dataset "SentiFive" includes 6,780 entries of নিশ্চিত নেতিবাচক (Strongly Negative), 6,537 entries of নিশ্চিত ইতিবাচক (Strongly Positive), 6,453 entries of নিরপেক্ষ (Neutral), 6,006 entries of কিছুটা নেতিবাচক (Slightly Negative), and 5,635 entries of নিশ্চিত ইতিবাচক (Strongly Positive), making it quite well balanced.

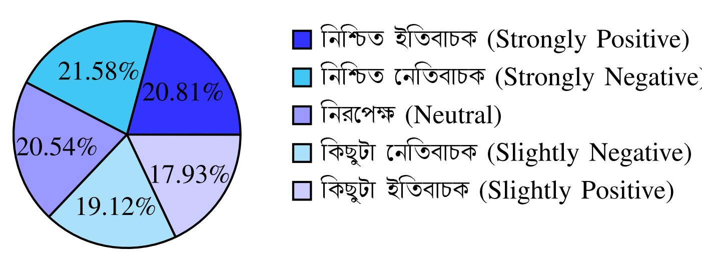

Fig. 1: Dataset Proportion to Each Class

2) Frequent Words: Table IV shows the top 10 most frequent words in each sentiment class along with their English translations. This gives a proper insight into the linguistic patterns and word usage associated with each sentiment.

Class	10 Most Frequent Words
নিশ্চিত ইতিবাচক (Strongly Positive)	অনেক (1031, Many), না (937, No), ভাই (937, Brother), ভালো (936, Good), আপনার (916, Your), জন্য (778, For), এই (755, This), ভিডিও (740, Video), করে (723, Do), আর (659, And)
কিছুটা ইতিবাচক (Slightly Positive)	না (1248, No), করে (799, Do), এই (741, This), ভাই (720, Brother), আর (710, And), কি (618, What), ভালো (608, Good), আপনার (531, Your), অনেক (508, Many), আমার (492, Mine)
নিরপেক্ষ (Neutral)	না (1325, No), কি (1052, What), ভাই (846, Brother), করে (775, Do), আর (629, And), এই (622, This), আমার (552, Mine), আমি (541, I), হবে (494, Will Be), থেকে (476, From)
কিছুটা নেতিবাচক (Slightly Negative)	না (1653, No), করে (889, Do), এই (790, This), কি (772, What), আর (753, And), ভাই (539, Brother), পুলিশ (510, Police), তো (466, So), হয় (460, Happens), হবে (402, Will Be)
নিশ্চিত নেতিবাচক (Strongly Negative)	না (1662, No), এই (1103, This), করে (1102, Do), আর (893, And), কি (889, What), পুলিশ (706, Police), সব (548, All), হবে (513, Will Be), কে (499, Who), হয় (470, Happens)

TABLE IV: Word Frequency in Each Class

IV. Feature Extraction

We applied Bag of Words (BoW) and TF-IDF on our ML models, whereas for DL models, we used Word2Vec word embeddings to capture semantic relationships among words.

A. Bag of Words (BoW)

BoW is an approach for extracting features from text data. It's very simple to understand and implement. Table V shows the BoW of the sentence "আমি বাংলায় মাতি উল্লাসে করি বাংলায় হাহাকার" (I talk, do rejoice, and lament in Bengali).

Word	Frequency
আমি (I), মাতি (talk), উল্লাসে (rejoice), করি (do), হাহাকার (lament)	1
বাংলায় (in Bengali)	2

TABLE V: Bag of Words of a Sentence

B. TF-IDF

TF means how many times a specific word is used in entire documents, and IDF means how important that word is in the entire document. So, TF and IDF make sense of the significance of the word throughout all the documents.

C. Word Embeddings

We implemented Word2Vec on tokenized text data using key parameters for DL models: a vector size of 100 for embedding dimension, a window size of 5 for capturing nearby word relationships, a minimum count of 3 to include all unique words. Text data was tokenized into integer sequences using a vocabulary, mapping each word to a unique token. Sequences were then padded for consistent input length.

V. Methodology for Sentiment Analysis

We applied both ML and DL algorithms to our dataset in order to establish the baseline performances that would show how different models could cope with the peculiarities of our dataset, such as five polarity classes and subtlety in the sentiment expressed in Bengali.

A. Machine Learning Model

1) Support Vector Classifier (SVC): SVC with a linear kernel was selected for its efficiency in high-dimensional feature spaces and its effectiveness in identifying linear decision boundaries for text classification. We set the regularization parameter to 1.0 to balance the trade-off between margin maximization and misclassification.

2) K-Nearest Neighbors (KNN): KNN is known for its simplicity and effectiveness in classification tasks, we tested different values of n (number of neighbors) to optimize performance, as smaller values can lead to sensitivity to noise. We experimented with $N = [0, 0.2, 0.6, 0.8, 1]$ to determine the optimal number of n.

3) Logistic Regression: It is widely used for binary classification, was evaluated using a range of regularization strengths $[0.001, 0.01, 0.1, 1, 10]$ to identify the best balance between underfitting and overfitting.

4) Multinomial Naive Bayes: It is well-suited for text classification with discrete features, was tested across various smoothing values $\alpha = [0, 0.2, 0.6, 0.8, 1]$, with optimal performance observed at higher smoothing levels.

5) Decision Tree: This classifier was implemented to model non-linear relationships by recursively splitting the dataset based on features that maximize information gain and minimize impurity.

B. Deep Learning Model

Deep learning (DL), a subset of ML, is more intricate and was developed to address the limitations of ML. We trained all the DL models using Adam as optimizer and using categorical cross-entropy loss and accuracy as the performance metric.

1) Convolutional Neural Network (CNN): We implemented our model using Keras for its simplicity and flexibility. A pre-trained Word2Vec-based Keras embedding layer was used to effectively capture semantic word associations. The model architecture included three convolutional blocks, each with two Conv1D layers (ReLU activation, kernel size 3, stride 1) and Dropout for regularization. MaxPooling1D or GlobalMaxPooling1D followed each block to reduce spatial dimensions. A dense layer with 200 ReLU-activated units and Dropout preceded the output layer, which used softmax activation over 5 units for multi-class classification.

2) Recurrent Neural Network (RNN): We implemented an RNN model with a simple RNN layer of 64 units to capture sequential patterns. This allowed us to compare the impact of different recurrent layers on classification accuracy.

3) Long Short-Term Memory (LSTM): This model uses a 64-unit LSTM layer to capture sequential dependencies in text. Two dense layers with 64 ReLU-activated units enhance feature learning, with Dropout layers in between to reduce overfitting. A softmax-activated output layer with 5 units enables multi-class classification. This architecture is well-suited for tasks requiring deeper analysis of sequential patterns.

4) BiLSTM: Based on [15], we implemented a BiLSTM model by extending our previous architectures. The key difference is the use of a 128-unit bidirectional LSTM, enabling the model to capture context from both past and future words. A softmax output layer with 5 units handles multi-class classification.

5) CNN-LSTM: We developed a hybrid model combining convolutional and recurrent layers to integrate local feature extraction with sequential pattern recognition. It begins with a Word2Vec embedding layer, followed by a Conv1D layer (200 filters, kernel size 3) to extract local features. A Bidirectional LSTM with 64 units captures temporal dependencies, followed by another BiLSTM layer for deeper contextual understanding. Dropout was added after the first LSTM to reduce overfitting. The model then passes through two dense layers (50 units, ReLU), a Flatten layer, and a final dense layer with 100 units and L2 regularization for better generalization. A softmax output layer with 5 units handles multi-class classification.

VI. Result Analysis

The dataset of 31,411 records was split into training and testing subsets, with 80% (25,129 entries) used for training and 20% (6,282 entries) for testing. To comprehensively evaluate model performance, experiments were conducted under three classification schemes: five-class, three-class, and two-class sentiment classification.

A. Five-Class Classification

The dataset comprised of five sentiment categories: strongly negative, slightly negative, neutral, slightly positive, and strongly positive. Deep learning models exhibited superior performance in this setting. Deep learning models, particularly LSTM and BiLSTM, outperformed others, each achieving 39% accuracy, followed by CNN and CNN-LSTM (38%) and RNN (37% and 36%). Traditional ML models scored below 35%, with TF-IDF variants of logistic regression and Naive Bayes slightly outperforming their BoW counterparts. Table VI shows F1 scores, with BiLSTM as the top performer.

Model	Accuracy	Macro F1	Weighted F1
Logistic Regression (BoW)	0.35	0.32	0.32
Multinomial NB (BoW)	0.34	0.31	0.31
KNN (BoW)	0.26	0.26	0.26
SVC (BoW)	0.33	0.31	0.32
Decision Tree (BoW)	0.26	0.26	0.26
Random Forest (BoW)	0.32	0.30	0.31
Logistic Regression (TF-IDF)	0.35	0.30	0.31
Multinomial NB (TF-IDF)	0.35	0.30	0.31
KNN (TF-IDF)	0.28	0.26	0.27
SVC (TF-IDF)	0.34	0.32	0.32
Decision Tree (TF-IDF)	0.27	0.26	0.27
Random Forest (TF-IDF)	0.34	0.31	0.32
LSTM	**0.39**	**0.34**	**0.35**
BiLSTM	0.39	0.33	0.36
RNN	0.36	0.33	0.34
CNN	0.38	0.32	0.34
CNN-LSTM	0.38	0.32	0.32

TABLE VI: Performance Metrics for Five-Class Sentiment Classification

Fig. 2: Confusion matrix for the five-class classification task using the LSTM model.

The confusion matrix shows strong correct predictions for classes 3 and 4. Most errors occur between neighboring sentiment levels, meaning the model struggles mainly with subtle polarity differences. This highlights the challenge of fine-grained sentiment classification.

B. Three-Class Classification

For the three-class configuration, strongly and slightly positive categories were merged into a single positive class, and strongly and slightly negative into a single negative class, while the neutral class was retained. Deep learning models again outperformed traditional methods, with BiLSTM and CNN-LSTM achieving approximately 59% accuracy. Among classical models, logistic regression with TF-IDF features reached 54%, surpassing its BoW variant. Detailed results are summarized in Table VII.

Model	Accuracy	Macro F1	Weighted F1
Logistic Regression (BoW)	0.53	0.43	0.49
Multinomial NB (BoW)	0.53	0.45	0.50
KNN (BoW)	0.43	0.41	0.43
SVC (BoW)	0.52	0.42	0.48
Decision Tree (BoW)	0.44	0.41	0.44
Random Forest (BoW)	0.52	0.46	0.50
Logistic Regression (TF-IDF)	0.54	0.45	0.51
Multinomial NB (TF-IDF)	0.53	0.41	0.48
KNN (TF-IDF)	0.46	0.41	0.45
SVC (TF-IDF)	0.53	0.42	0.49
Decision Tree (TF-IDF)	0.44	0.40	0.43
Random Forest (TF-IDF)	0.53	0.45	0.50
LSTM	0.58	0.43	0.52
BiLSTM	**0.59**	**0.52**	**0.57**
RNN	0.56	0.42	0.50
CNN	0.58	0.43	0.51
CNN-LSTM	0.59	0.52	0.56

TABLE VII: Performance Metrics for Three-Class Sentiment Classification

C. Two-Class Classification

The binary classification setup excluded the neutral class and consolidated all positive and negative sentiments into two categories. This simplification resulted in the highest accuracies across all models. LSTM achieved the best accuracy at 74%, followed by BiLSTM and CNN close behind at approximately 73%. Among traditional classifiers, SVC with

TF-IDF attained the highest accuracy of 67%. Table VIII details the performance metrics for this classification task.

Model	Accuracy	Macro F1	Weighted F1
Logistic Regression (BoW)	0.66	0.66	0.66
Multinomial NB (BoW)	0.66	0.66	0.66
KNN (BoW)	0.60	0.59	0.59
SVC (BoW)	0.65	0.65	0.65
Decision Tree (BoW)	0.59	0.59	0.59
Random Forest (BoW)	0.66	0.66	0.66
Logistic Regression (TF-IDF)	0.66	0.66	0.66
Multinomial NB (TF-IDF)	0.66	0.66	0.66
KNN (TF-IDF)	0.60	0.57	0.57
SVC (TF-IDF)	0.67	0.67	0.67
Decision Tree (TF-IDF)	0.60	0.60	0.60
Random Forest (TF-IDF)	0.66	0.65	0.65
LSTM	**0.74**	**0.74**	**0.74**
BiLSTM	0.73	0.73	0.73
RNN	0.70	0.70	0.70
CNN	0.73	0.73	0.73
CNN-LSTM	0.72	0.72	0.72

TABLE VIII: Performance Metrics for Two-Class Sentiment Classification

D. Overall Insights

1) Accuracy improves as the number of sentiment classes decreases. Simplifying the classification task reduces ambiguity and complexity, leading to better insight.
2) Deep learning models consistently outperform traditional machine learning algorithms. Architectures incorporating LSTM units are particularly effective in capturing contextual dependencies.
3) Certain ML algorithms, such as KNN and Decision Trees, show limited effectiveness, especially for the five-class task, due to challenges with high-dimensions.
4) TF-IDF feature representation generally yields better performance than Bag-of-Words, thanks to its superior ability to weigh informative terms.
5) Binary sentiment classification yields the highest accuracies across all models, underscoring the value of the task simplification in practical applications.

VII. Conclusion

This research addresses the shortage of standard Bengali sentiment analysis resources by introducing **SentiFive**, a robust and a diverse corpus of 31,411 Bengali YouTube comments annotated across five sentiment categories. Unlike prior datasets limited to binary or ternary sentiment and labels, SentiFive enables finer-grained sentiment modeling, offering greater insight into user opinions which is particularly important for low-resource languages like Bengali. Although performance declines as the number of classes increases, deep learning models demonstrated strong potential in handling this complexity. The multiclass dataset holds promise for advancing applications such as opinion mining and sentiment-aware social media analysis. As part of our future work, we aim to explore advanced modeling strategies to address performance challenges in multi-class classification. To encourage further research in Bengali NLP, the dataset, accompanying resources, and related codebase will be made publicly available.

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